

Ankit Kumar

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Professional Summary

Experienced Applied Scientist with a background in finance and tech industries, having worked at Rivos, Amazon, Morgan Stanley and Olacabs, eager to leverage my academic expertise and industry experience to implement formal methods in practical applications, driving innovation and enhancing software quality in real-world projects.

Computer Skills

C, C++, Java, Rust, Lisp, ACL2, ACL2 Sedan, Coq, Haskell, Emacs, Git, L^AT_EX

Professional Experience

Rivos Inc.

Portland, OR, USA

Member of Technical Staff, Intern

Summer 2023

- Formalized and proved security properties about the page walk algorithm used by a RISC-V based hypervisor (high level model).
- Reduced ($\approx 10\times$) the time required to prove existing security theorems stated in Jaspergold (low level model) from days to hours, by transferring some of the identified helpful lemma from the high level model.
- ACL2, ACL2 Sedan, Jaspergold

Amazon

Minneapolis, MN, USA

Applied Scientist Intern

Summer 2022

- Created a tool to identify possible critical data leaks from AWS resources.
- Demonstrated the use of taint-analysis tools for cloud level tracking of critical data across AWS resources.
- Patented this technique, and received an incline offer from Amazon.
- Java, Grails, github, shell scripting

Olacabs

Bengaluru, KA, India

Software Development Engineer II

2012-2014

- Wrote the first version of the Olacabs (Indian ride-hailing app service, like Uber) android consumer app, responsible for over 80% of all cab bookings.
- Debugged the Olacabs driver app, reducing instances of phone crashes by more than 60%.
- Worked on cashless payment and online wallet systems.
- Java, Android, Git, Ruby on Rails

Morgan Stanley Advantage Services

Mumbai, MH, India

Associate

2010-2012

- Refactored a derivatives trading platform to be region aware.
- Rolled out the platform to new markets (HK, SG) which increased the number of trades it handled by more than 2x.

- .NET, Java, Spring, Perforce

Education

Northeastern University

Boston, MA, USA

PhD in Computer Science

2018–2025

Applied theorem proving, counter-example generation and the theory of refinement to analyze the Gossipsub P2P protocol. Also worked on automated feedback for homework submissions, proof checking and decision procedures for solving string equations.

Thesis: Refinement based reasoning of P2P protocols is feasible and useful

IIT Kanpur

Kanpur, UP, India

M.Tech in Computer Science

2015–2017

IIT (BHU) Varanasi

Varanasi, UP, India

B.Tech in Electrical Engineering

2006–2010

Publications

Ankit Kumar, Max von Hippel, Panagiotis Manolios & Cristina Nita-Rotaru (2023): *Verification of GossipSub in ACL2s*. In: *Proceedings of the 18th International Workshop on the ACL2 Theorem Prover and Its Applications*, doi:10.4204/EPTCS.393.10.

Ankit Kumar, Max von Hippel, Panagiotis Manolios & Cristina Nita-Rotaru (2024): *Formal Model-Driven Analysis of Resilience of GossipSub to Attacks from Misbehaving Peers*. In: *2024 IEEE Symposium on Security and Privacy (SP)*, doi:10.1109/SP54263.2024.00017.

Ankit Kumar & Panagiotis Manolios (2021): *Mathematical Programming Modulo Strings*. In: *2021 Formal Methods in Computer Aided Design (FMCAD)*, doi:10.34727/2021/isbn.978-3-85448-046-4.

Ankit Kumar, Andrew Walter & Panagiotis Manolios (2023): *Automated Grading of Automata with ACL2s*. *Electronic Proceedings in Theoretical Computer Science*, doi:10.4204/eptcs.375.7.

Andrew T. Walter, Ankit Kumar & Panagiotis Manolios (2023): *Calculational Proofs in ACL2s*. CoRR abs/2307.12224, doi:10.48550/ARXIV.2307.12224. arXiv:2307.12224.

Andrew T. Walter, Ankit Kumar & Panagiotis Manolios (2023): *Proving Calculational Proofs Correct*. In: *Proceedings of the 18th International Workshop on the ACL2 Theorem Prover and Its Applications*, doi:10.4204/EPTCS.393.11. Available at <https://doi.org/10.4204/EPTCS.393.11>.

Awards and Honors

Award: Best Student Paper Award for our paper “A Formalization of the Correctness of the Floodsub Protocol” at ACL2 Workshop 2025.

Fellowship: Dissertation Completion Fellowship during Summer 2024.

Invited Talk - Syracuse University 2024: Formal Model-Driven Analysis of Resilience of GossipSub to Attacks from Misbehaving Peers

Award: Best Student Paper Award for our paper “Verification of GossipSub in ACL2s” at ACL2 Workshop 2023.

Invited Talk - IPFS Camp at Lisbon, 2022: Formal Analysis of GossipSub : Property Driven Dev for Security

Fellowship: Student Travel Fellowship to attend PLMW at POPL 2019 conference.

Award: First Prize at Droidcon India - Bangalore 2013 Hackathon